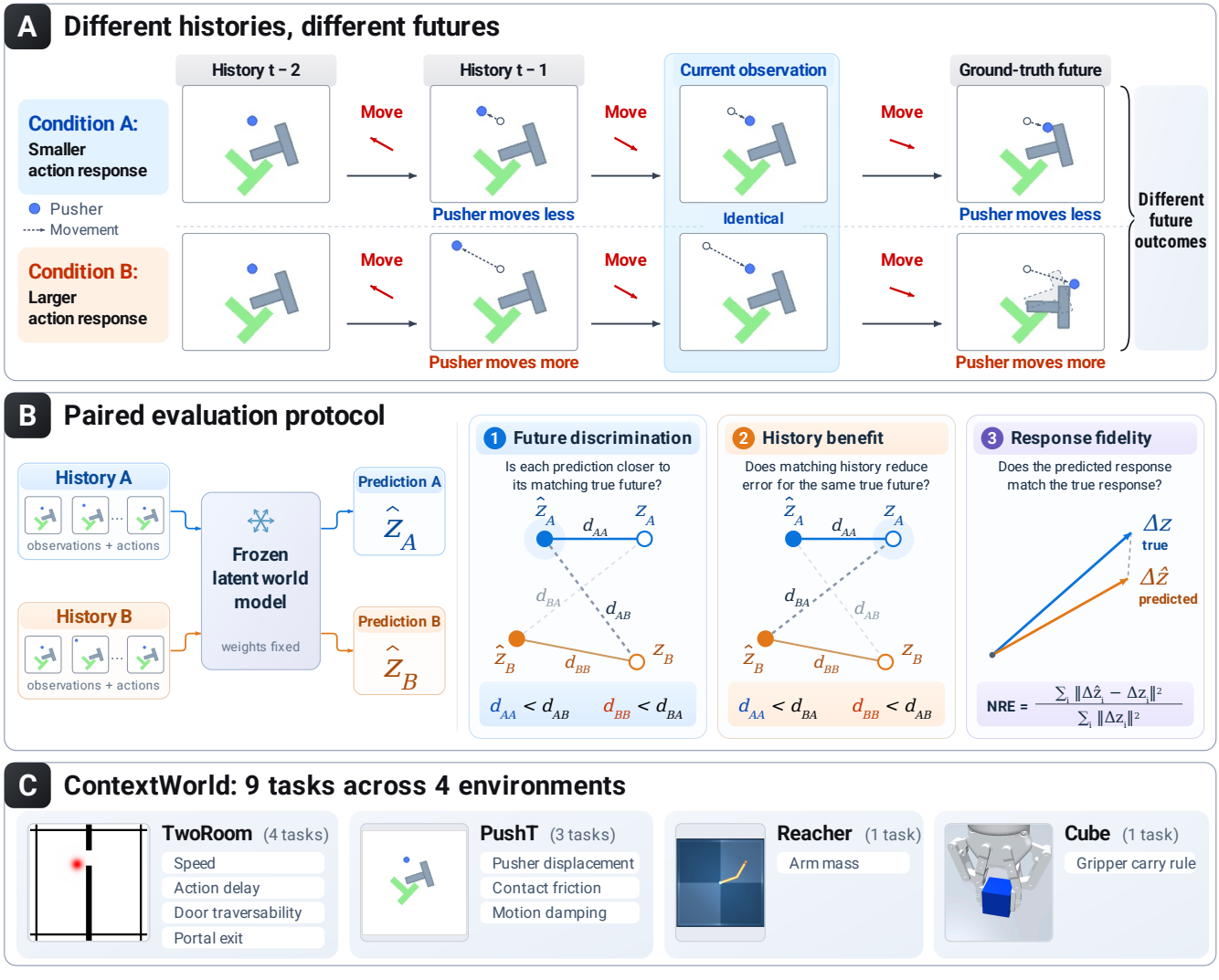

A
Different histories, different futures
History t − 2
History t − 1
Current observation
Ground-truth future
Condition A:
Smaller action response
Pusher
Movement
Condition B:
Larger action response
Move
Pusher moves less
Identical
Pusher moves less
Pusher moves more
Pusher moves more
Different future outcomes
B
Paired evaluation protocol
History A
observations + actions
History B
observations + actions
Frozen latent world model
weights fixed
Prediction A
Prediction B
1 Future discrimination
Is each prediction closer to its matching true future?
2 History benefit
Does matching history reduce error for the same true future?
3 Response fidelity
Does the predicted response match the true response?
true
predicted
C
ContextWorld: 9 tasks across 4 environments
TwoRoom (4 tasks)
Speed
Action delay
Door traversability
Portal exit
PushT (3 tasks)
Pusher displacement
Contact friction
Motion damping
Reacher (1 task)
Arm mass
Cube (1 task)
Gripper carry rule